

Le Chatelier's Principle

Answer key with worked solutions

A. Understanding

1. Le Chatelier's principle.

If a system at equilibrium is disturbed by a change in concentration, pressure or temperature, the position of equilibrium shifts in the direction that **partly counteracts** the change.

B. The Haber process

2. Predictions.

(a) Shifts **right**: more NH_3 ; K unchanged. (b) Shifts **right**: more NH_3 forms; K unchanged. (c) Shifts **right** ($4 \text{ mol gas} \rightarrow 2 \text{ mol gas}$): more NH_3 ; K unchanged. (d) Forward reaction is exothermic, so it shifts **left**: less NH_3 ; **K decreases**. (e) **No shift** and K unchanged; equilibrium is reached faster.

3. Why 450°C .

At low temperatures the equilibrium yield is higher, but the **rate** is far too slow, even with a catalyst. 450°C is a **compromise**: a reasonable yield produced quickly enough. (High pressure and recycling of unreacted gases also improve the overall yield.)

C. Colour and temperature

4. Cooling.

The forward reaction is endothermic ($\Delta H = +57 \text{ kJ}$), so cooling favours the **exothermic reverse** reaction. More colourless N_2O_4 forms, and the mixture becomes **paler**.

5. Compression.

Halving the volume immediately **doubles all concentrations**, so the brown colour first **darkens**. The higher pressure then shifts the equilibrium toward **fewer moles of gas** ($2 \text{ mol NO}_2 \rightarrow 1 \text{ mol N}_2\text{O}_4$), so some NO_2 is used up and the colour becomes slightly **paler**. The change is only partly counteracted, so it stays darker than at the start.

D. Special cases

6. HI equilibrium.

There are **2 mol of gas on each side**, so neither direction reduces the number of gas molecules. Changing the pressure changes all concentrations by the same factor, and Q stays equal to K .

7. Adding CaCO_3 .

No shift. CaCO_3 is a pure solid, so it does not appear in $K_c = [\text{CO}_2]$; adding more does not change Q .

E. Using Q

8. Q after adding H_2 .

New $[\text{H}_2] = 0.0222 \text{ M} + 0.100 \text{ M} = 0.122 \text{ M}$

$$Q = \frac{(0.156 \text{ M})^2}{(0.122 \text{ M})(0.0222 \text{ M})} = 8.99$$

The units of M^2 cancel. $Q = 8.99 < K_c = 49.0$, so the reaction goes **forward** (right).

F. Challenge

9. New equilibrium.

Let x (in M) be the decrease in $[\text{I}_2]$. New equilibrium: $[\text{H}_2] = 0.12222 \text{ M} - x$, $[\text{I}_2] = 0.02222 \text{ M} - x$, $[\text{HI}] = 0.15556 \text{ M} + 2x$. Working in units of M:

$$49.0 = \frac{(0.15556 + 2x)^2}{(0.12222 - x)(0.02222 - x)}$$

Expanding and collecting terms:

$$45.0x^2 - 7.700x + 0.1089 = 0$$

$$x = \frac{7.700 - \sqrt{(7.700)^2 - 4(45.0)(0.1089)}}{2(45.0)} = \frac{7.700 - 6.300}{90.0} = 0.01556$$

(The other root, 0.1556, would make $[\text{I}_2]$ negative, so it is rejected.)

$[\text{H}_2] = 0.12222 \text{ M} - 0.01556 \text{ M} = \mathbf{0.107 \text{ M}}$; $[\text{I}_2] = 0.02222 \text{ M} - 0.01556 \text{ M} = \mathbf{0.00667 \text{ M}}$;

$[\text{HI}] = 0.15556 \text{ M} + 0.03111 \text{ M} = \mathbf{0.187 \text{ M}}$

Check: $\frac{(0.18667)^2}{(0.10667)(0.00667)} = 49.0 \checkmark$